

Project Report

Multiphysics Modelling using COMSOL ME46095

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1 Introduction

This report describes a straightforward method of cooling a battery cell's tabs with air.

No energy source is perfectly efficient. This means that in addition to providing useful energy, it generates parasitic heat. Lithium-ion Batteries suffer especially from this problem, as the temperature limits for their safe operation are quite severe - almost all chemistries must not exceed a maximum temperature of $T_{\max} = 60^{\circ}\text{C}$, otherwise harmful irreversible chemical reactions start occurring in the electrodes.

Therefore, to be able to use a battery as an effective energy source with high power demands, it needs to be cooled to ensure enough heat is extracted to meet the maximum temperature demand.

This simulation was conducted with a specific case in mind: a traction accumulator for a Formula Student vehicle. There are many approaches to cool the battery down, but the accumulator is built to be as compact as possible. Therefore, the air-cooled concept was chosen, as it allows for the most compact design.

Simulating the entire accumulator with 270 battery cells would be too demanding computationally and memory-wise¹. Therefore, only the simplest case - a single battery in free stream of air - was simulated.

The aim of this simulation is to determine the optimal airspeed sufficient to keep the battery under the maximum temperature for 20 minutes, which roughly corresponds with the longest race the vehicle is designed for.

2 Model Definition

The situation is very simple: a single pouch-cell battery with its tabs protruding upwards. The goal is to cool the active material of the cell, which is contained within the pouch. However, the pouch casing is generally a poor thermal conductor, decreasing the cooling effectivity, if the air were to blow over it. Instead, only the tabs will be cooled, as they are made of copper - a very good electrical and thermal conductor - and are almost ideally connected to the electrodes, which need to be cooled.

2.1 Geometry

There are two parts to the geometry of this problem: the wind tunnel, within which the air will flow, and the battery cell itself.

The battery is modeled after the used cell. It is a pouch cell, within which the electrodes are tightly wound in many layers. The layered structure will be modeled with the material properties, so the geometry is very simple: a prism containing the active battery material with thin copper current collectors (tab connections) on either side. Copper tabs protrude upwards into the wind tunnel

¹A periodic condition cannot be applied to the problem, as the air flow field is not periodic, but rather chaotic - it is not the same for every battery cell.

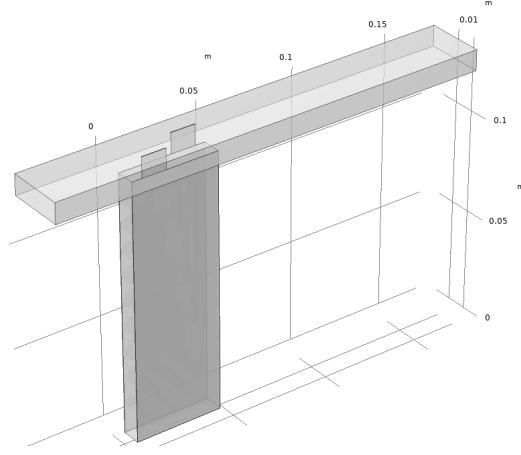


Figure 1: The simulated geometry - pouch cell with tabs and the wind tunnel above

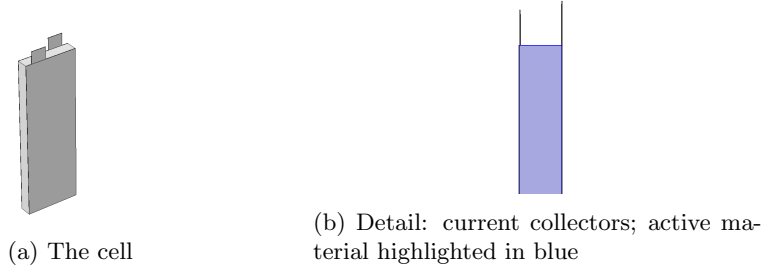


Figure 2: Cell geometry

2.2 Material and Boundary Conditions

2.2.1 Materials

The coolant is air with modeled as ideal gas.

The battery domain uses copper for the tabs and a custom material for the active region. The electrodes are tightly wound inside the pouch, which results in through-plane and in-plane thermal conductivities being vastly different. A nonisotropic thermal conductivity is used for the custom material given as

$$k = \begin{pmatrix} k_{\text{in}} & 0 & 0 \\ 0 & k_{\text{through}} & 0 \\ 0 & 0 & k_{\text{in}} \end{pmatrix} = \begin{pmatrix} 29.56 & 0 & 0 \\ 0 & 0.897 & 0 \\ 0 & 0 & 29.56 \end{pmatrix} [\text{Wm}^{-1}\text{K}^{-1}]$$

Domain	Dimension	Value [mm]	Comment
Cell	Pouch width	43	
	Pouch height	125.5	
	Active mat. thickness	9.1	
	Current coll. thickness	0.2	Same as tab thickness
	Total thickness	9.5	
	Tab length	10	
	Tab width	12	
	Tab thickness	0.2	
Wind tunnel	Height	10	Same as tab length
	Width	28.5	3x pouch thickness
	Length	215	5x pouch width

Figure 3: Dimensions of the geometry

These values are typical for the LiCoO_2 chemistry.

Material	ρ [kg m^{-3}]	c_p [$\text{J kg}^{-1}\text{K}^{-1}$]	k [$\text{W m}^{-1}\text{K}^{-1}$]
Air	1.225	1005	0,026
Copper	8960	385	401

Figure 4: Built-in library material properties

2.2.2 Airflow

The most simple CFD simulation requires two types of boundary conditions. First, conditions that force the air to move:

1. A *velocity inlet* determines the airspeed of the incoming air. The air will enter the wind tunnel through one of it's walls. As determining the optimal velocity is the goal of this study, the velocity will be swepted. The original inlet face is the one facing the direction of the airflow. However, the boundary condition is applied to adjacent faces as well (all except the outlet and cell tabs). Doing this forces the velocity on these faces to be exactly the same as the original inlet velocity. Thus, free stream will be simulated and no boundary layers will form on the sides of the wind tunnel.
2. A *pressure outlet* allows the air to exit the wind tunnel. This condition forces a constant static pressure along the wall and allows non-zero velocity there. To simulate free stream airflow, this boundary ocndision is applied

to the wall opposite to the original inlet. The gauge pressure at the outlet is set to zero, which corresponds to atmospheric pressure.

Second, the *wall* boundary conditions constrain the airflow to the geometry by forcing tangent flow and zero velocity on the wall. This will be applied to every other wall of the geometry (that is not inlet or outlet).

As the problem deals with heat transfer, an ambient temperature is set to the inlet.

Condition	Value	
Ambient temperature	25 °C	Applied at inlet
Pressure outlet	0 Pa	Gauge pressure
Walls	No slip	
Initial velocity	Inlet velocity	Initial value for transient simulation

Figure 5: Boundary and initial conditions

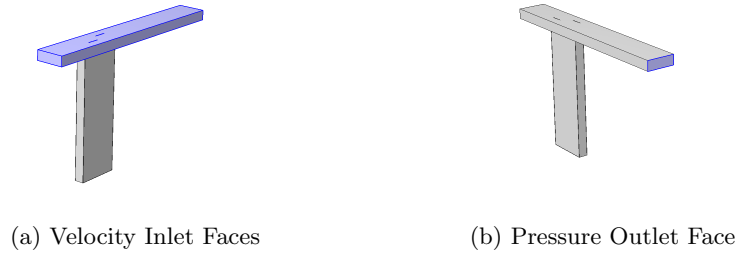


Figure 6: Airflow Boundary Conditions

2.2.3 Battery

The battery parameters are based on the cell used in the actual accumulator.

Property	Value	Unit	
Maximum voltage	4.2	V	
Internal resistance	3	m Ω	
Applied current	-18	A	Discharge current
Capacity	7000	mAh	

The current flowing through the battery is empirically measured from gathered data from previous races. In reality, the current is not constant. But for the sake of this simulation, it will be simulated as a constant current, which generates an equivalent amount of waste heat through Joule heating.

The COMSOL module for battery physics requires an ohmic overpotential

$$\eta_{\text{IR}} = R_{\text{int}}I = 0.003 \cdot 18 = 0.0054 \text{ V},$$

which is really a voltage drop induced by the applied current across the internal resistance, and an activation overpotential, which describes a threshold voltage for the chemical reactions to occur. A common dimensionless value for LiCoO_2 cells is

$$J_0 = 0.85.$$

The applied discharge current $I = 18\text{A}$ corresponds to the discharge rate of 3C.

$$\text{Discharge rate} = \frac{I}{C} = \frac{18}{7} = 1.57\text{C}. \quad (1)$$

This means that all of the cell's energy will be discharged in $\frac{60}{2.57} = 23.33$ minutes, which is sufficient for the simulation goal.

2.3 Physics Coupling

Given the nature of the problem, there will be coupling between multiple physics. Most notably between fluid flow and heat transfer, but also between electrochemical processes and heat generation.

The current passing through the battery cell will generate heat, which can be thought of as energy loss on the cell's internal resistance. This waste heat will propagate through the solid material and into the fluid. The rate at which the heat is being exchanged with the fluid depends heavily on the local flow conditions, but mainly on the thickness of the boundary layer and the airflow velocity.

The system is governed by the following equations:

1. Battery energy

$$E_{\text{cell}} = E_{\text{OCV}}(\text{SoC}, t) + \eta_{\text{IR}} + \eta_{\text{act}}, \quad (2)$$

where η_{IR} is the ohmic overpotential and η_{act} is the activation overpotential.

2. Volumetric Joule heating

$$\dot{Q} = \mathbf{J} \cdot \mathbf{E}, \quad (3)$$

where $\mathbf{J}$ is the current density and $\mathbf{E}$ is the electric field.

3. Laminar flow

$$\rho \frac{\partial \mathbf{u}}{\partial t} + \rho(\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla \cdot (-p\mathbf{l} + \mathbf{K}) + \mathbf{F}, \quad (4)$$

$$\rho \nabla \cdot \mathbf{u} = 0 \quad (5)$$

where $\mathbf{u}$ and ρ are the fluid's speed and density.

4. Heat transfer

$$\rho c_p \frac{\partial T}{\partial t} + \rho c_p \mathbf{u} \cdot \nabla T + \nabla \cdot \mathbf{q} = Q + Q_{\text{ted}} \quad (6)$$

$$\mathbf{q} = -k \nabla T, \quad (7)$$

where ρ and c_p are the material density and specific heat, T is the temperature and $\mathbf{q}$ is the heat flow.

3 Implementation in COMSOL

Aside from the 3D airflow simulation, a dedicated battery module is used to ensure correct heat generation.

3.1 Battery Cell Heating

For simulation of the battery heating, the "Lumped Battery" module is used. This module essentially calculates the Joule heating based on the cell's equivalent circuit (most importantly the internal resistance) and applies it as a volumetric heat generation within the cell domain.

This module uses time-dependent equations. This means that steady-state simulations cannot be done. Luckily, the simulation goal involves a time-dependent simulation, but it will take longer to compute.

3.2 Air Cooling

Laminar flow solver is used for the airflow. We are interested in what is happening around the cell tabs. The flow will be laminar at Reynolds numbers lower than 2300. The maximum velocity for the flow to remain laminar can be expressed as

$$v_{\text{laminar}}^{\text{max}} = \frac{\mu \text{Re}}{\rho w_{\text{tab}}} = \frac{1.85 \cdot 10^{-5} \cdot 2300}{1.225 \cdot 0.012} = 2.89 \text{ ms}^{-1}, \quad (8)$$

where the tab width w_{tab} is the reference length scale for the Reynolds number. The velocities are not expected to exceed $v_{\text{laminar}}^{\text{max}}$, though, so unless the simulation shows otherwise, the laminar solver will suffice.

4 Model Validation

Using the formula for heat exchange with a closed volume of fluid

$$\dot{Q} = \dot{m}c_p\Delta T, \quad (9)$$

we can validate the heat generation within the cell and the heat exchange with air. $\dot{Q}$ refers to the total heat generation rate accepted by the fluid, which results in the temperature rise ΔT . $\dot{m}$ is the mass flow and c_p is isobaric specific heat of the fluid.

Rewriting the formula to obtain temperature change yields

$$\Delta T = \frac{\dot{Q}}{c_p\dot{m}}. \quad (10)$$

Using this, compare the theoretical and simulated temperature change.

In a model case of the simulation with inlet air velocity set to $v = 2 \text{ ms}^{-1}$, we take

$$\dot{Q} = R_{\text{int}}I^2 = 0.003 \cdot 18^2 = 0.972 \text{ W} \quad (11)$$

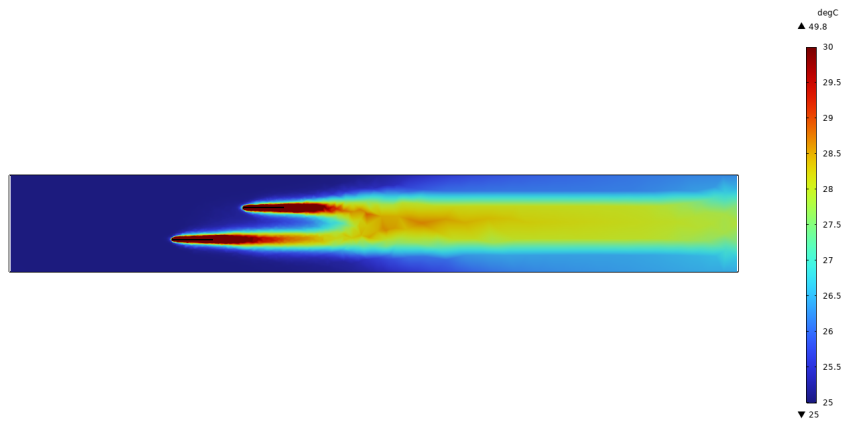
and

$$\dot{m} = \rho Av = 1.225 \cdot (0.01 \cdot 0.0285) \cdot 2 = 0.00069825 \text{ kg s}^{-1}, \quad (12)$$

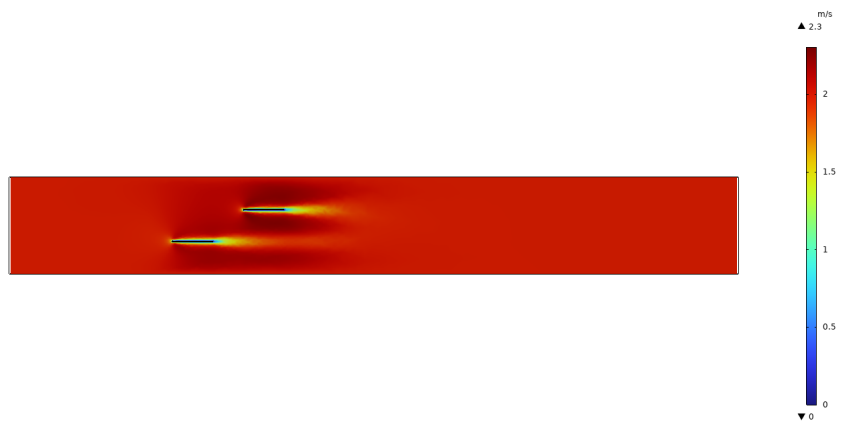
we obtain expected temperature change

$$\Delta T_{\text{exp}} = \frac{0.972}{1005 \cdot 0.00069825} = 1.385 \text{ K} \quad (13)$$

The temperature rose by $\Delta T = 2.165 \text{ K}$, which is nearly twice the expected value. However, formula 9 only considers a perfectly mixed fluid stream, where the heat is transferred into its entire volume. As is apparent from figure 7a, only about half of the air stream actually participates in the heat transfer. The theoretical ΔT is therefore underestimated by roughly a factor of 2, which correlates to the validation result.



(a) Air temperature



(b) Air velocity

Figure 7: Model case for validation

5 Results

To obtain the optimal velocity, a velocity sweep is conducted. The first attempt considers 3 velocities: 1, 2, 3 ms^{-1} .

As is apparent from figure 8b, the ideal velocity lies somewhere between 1 and 2 ms^{-1} . The next sweep was done in this interval with increment 0.25 ms^{-1} . The optimal inlet velocity seems to be

$$v = 1.25 \text{ ms}^{-1}, \quad (14)$$

where after 20 minutes of loading, the maximum temperature inside the battery cell is

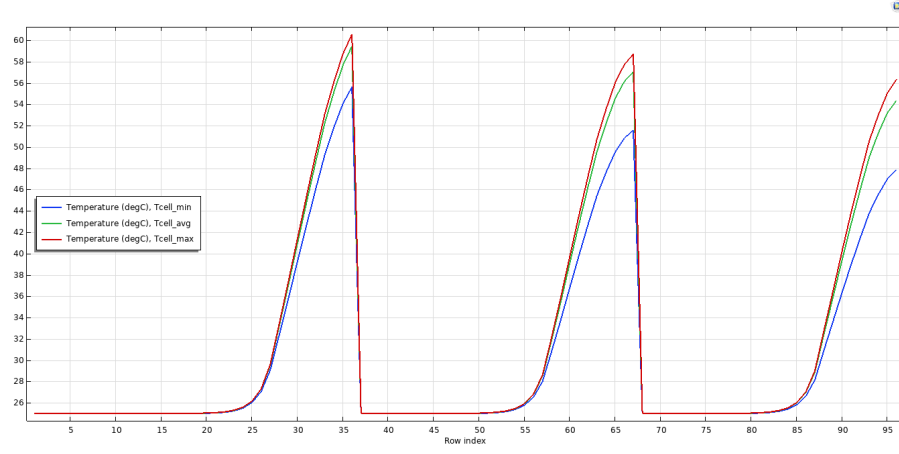
$$T_{\text{max}} = 59.93^\circ\text{C}. \quad (15)$$

6 Conclusion

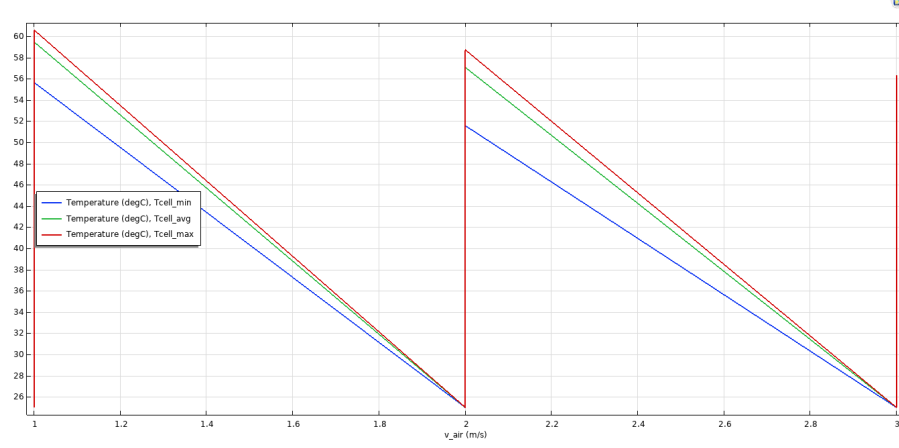
This report covers a study of an air-cooled battery cell used in a Formula Student accumulator pack. The requirements and ambient conditions were set based on empirical data from the competition.

The study involves simulation of the heating of the battery, as well as simulation of the heat being transferred to the coolant. The geometry is entirely created in the COMSOL environment with minimal simplifications.

In the study, an optimal inlet velocity was determined, which would ensure the battery doesn't exceed its thermal limit for the duration of the most demanding race. This result can be used in further design of the entire pack.

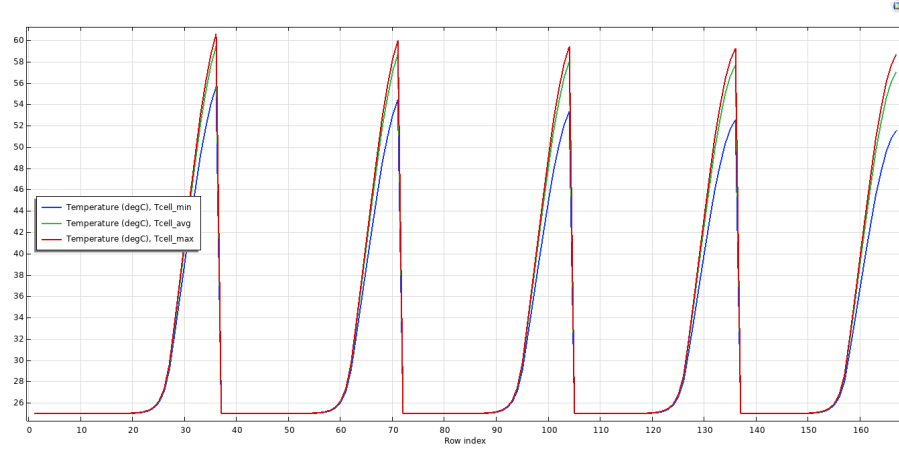


(a) x axis: index

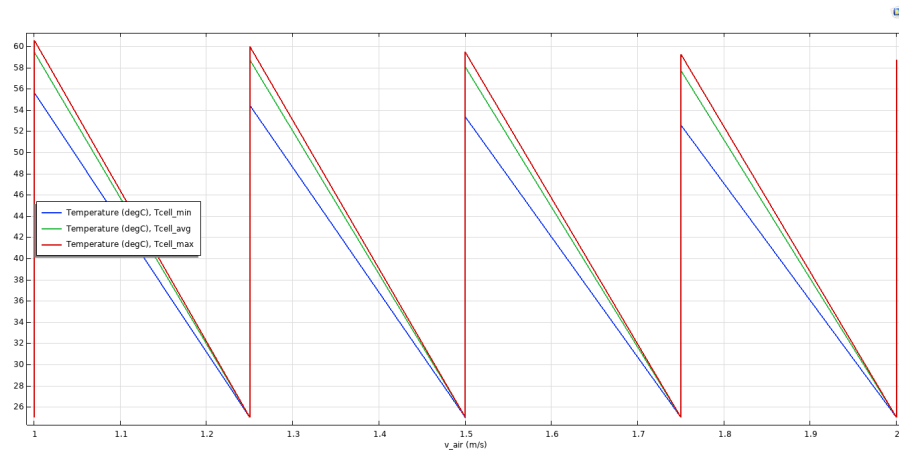


(b) x axis: inlet air velocity

Figure 8: Velocity sweep $v \in (1, 1.25, 1.5, 1.75, 2)$

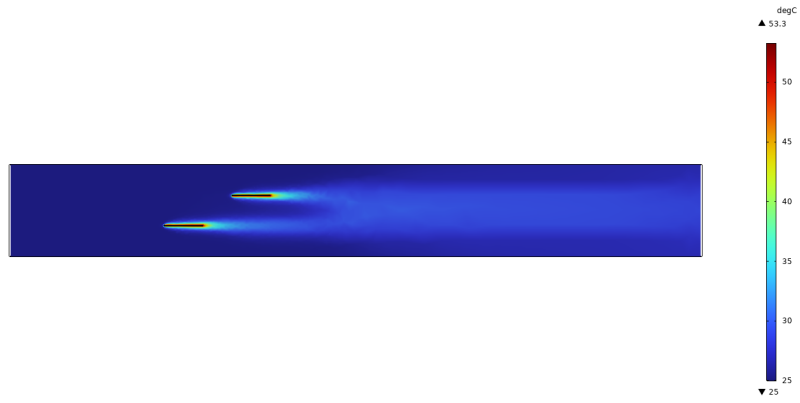


(a) x axis: index

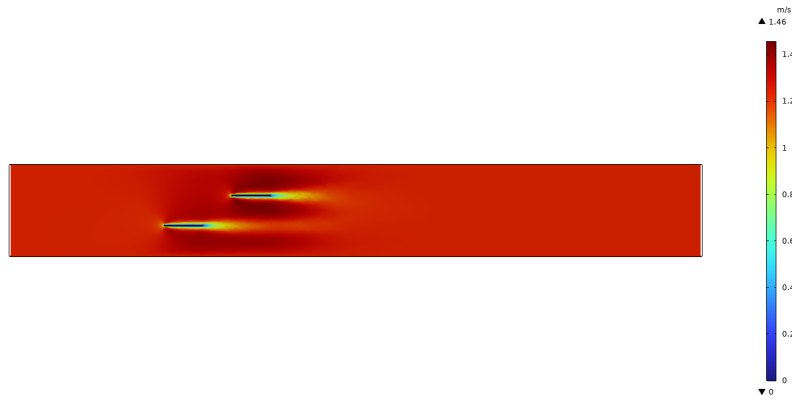


(b) x axis: inlet air velocity

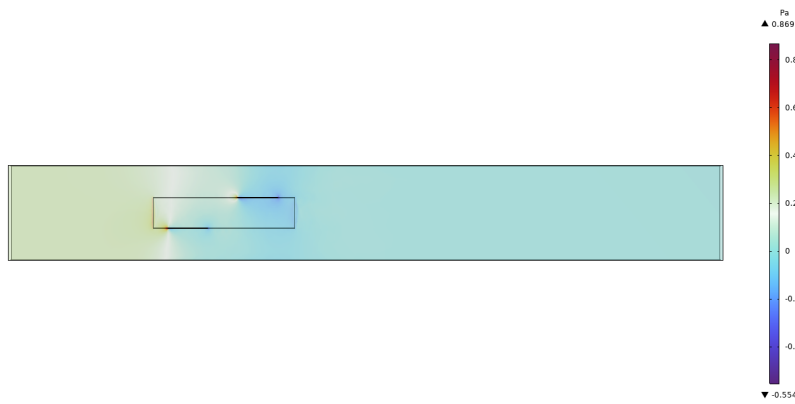
Figure 9: Velocity sweep $v \in (1, 2, 3)$



(a) Air temperature

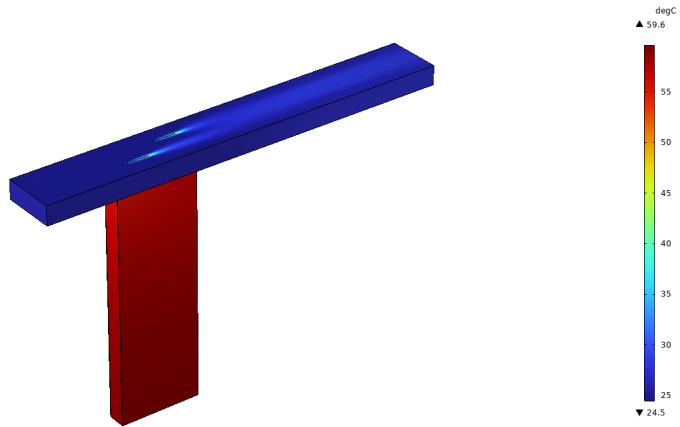


(b) Air velocity

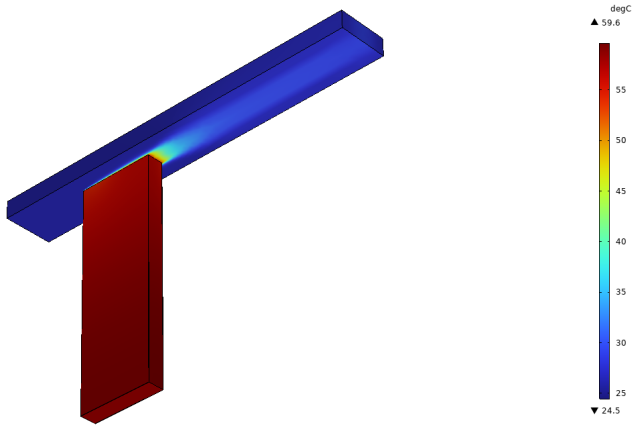


(c) Air static pressure

Figure 10: Final simulation: $v_{\text{in}} = 2 \text{ ms}^{-1}$



(a) Top view



(b) Bottom view

Figure 11: Final simulation: temperature

7 Appendix: Modeling in COMSOL Multiphysics

7.1 Model Wizard

Set up the project environment.

1. Select a 3D study.
2. Select Electrochemistry - Batteries - Lumped Battery.
3. Select Fluid Flow - Single-Phase Flow - Laminar Flow.
4. Select Heat Transfer - Heat Transfer in Solids and Fluids.
5. Select a time dependent study.
6. In the Model Builder, right-click the Multiphysics tab and add Electrochemical Heating and Nonisothermal flow.

7.1.1 Parameters

Define the basic parameters

1. In the Model Builder, right-click Global Definitions and add Parameters
2. Rename the new parameter category and define all needed parameters (it is beneficial to group parameters according to their usage)
3. It is important to define a parameter for the inlet velocity, denote it v_{air} .

7.1.2 Geometry

All used geometry is very simple. The geometry consists of 6 blocks: 1 active cell region, 2 current collectors, 2 cell tabs and 1 wind tunnel.

1. Right-click Geometry 1 and add Block.
2. Set the desired size and position in the menu.
3. Do the same for all other blocks.
4. Position the current collectors on either side of the active region.
5. Position the tabs atop the current collectors.
6. Position the wind tunnel atop the cell, intersecting the tabs.

h_cell	125.5 [mm]	0.1255 m	
w_cell	43 [mm]	0.043 m	
t_cell	9.5 [mm]	0.0095 m	
t_tab	0.2 [mm]	2E-4 m	
h_tab	10 [mm]	0.01 m	
w_tab	12 [mm]	0.012 m	
tab_offset	5 [mm]	0.005 m	From edge

(a) Cell Dimensions

Vnom	3.7 [V]	3.7 V	Nominal voltage
Vmax	4.2 [V]	4.2 V	Charge cutoff voltage
Vmin	2.5 [V]	2.5 V	Discharge cutoff voltage
Capacity	7 [Ah]	25200 C	Nominal capacity
Iavg	-18 [A]	-18 A	Mean current through 1 cell
Rint	0.003 [ohm]	0.003 Ω	Cell internal resistance in Ohms
eta_ohmic	abs(Iavg*Rint)	0.054 V	Ohmic overpotential at Iavg
J0	0.85	0.85	Activation overpotential
SOCinit	1	1	Initial SoC

(b) Cell Electrical Properties

kT_batt_ang	(kT_pos*L_pos+kT_neg*L_neg)	29.557 W/(m·K)	Battery thermal conductivity, angular
kT_batt_r	L_batt/(L_pos/kT_pos+L_neg/kT_neg)	0.89724 W/(m·K)	Battery thermal conductivity, radial
rho_batt	(rho_pos*L_pos+rho_neg*L_neg)/L_batt	2055.2 kg/m ³	Battery density
Cp_batt	(Cp_pos*L_pos+Cp_neg*L_neg)/L_batt	1169.1 J/(kg·K)	Battery heat capacity

(c) Cell Thermal Properties

Tamb	25 [degC]	298.15 K	Ambient temperature
Tmax_cell	60 [degC]	333.15 K	Maximum allowed cell temperature

(d) General Thermal Properties

h_wt	h_tab	0.01 m	Height
w_wt	5*w_cell	0.215 m	Length
l_wt	3*t_cell	0.0285 m	Width
v_air	1.25 [m/s]	1.25 m/s	Airflow velocity
T_air	Tamb	298.15 K	Air temperature
p_ref	101325 [Pa]	1.0133E5 Pa	Reference pressure

(e) Wind Tunnel Dimensions and Flow Properties

Figure 12: Defined parameters (some are unused)

7.1.3 Physics Domains

The created geometry needs to be assigned to the used physics modules.

1. Select Lumped Battery, in the Domain Selection select the block corresponding with the active cell material.
2. Select Laminar Flow, , in the Domain Selection select the block corresponding with the wind tunnel.
3. Select Heat Transfer in Solids and Fluids, in the Domain Selection select all blocks that are not the wind tunnel.

7.1.4 Materials

Now define the materials. Air and copper are easily added from the built-in library, but the active material of the cell must be custom-defined.

1. Right-click Materials - Add Material from Library.
2. In the Built-In tab, find Air and Copper and add them.
3. Right-click Heat Transfer in Solids and Fluids, add a solid domain (Solid) and rename it accordingly.
4. In the Heat Conduction, Solid tab, select Thermal Conductivity: User defined and fill the k matrix with your parameters.
5. In the Thermodynamics, Solid tab, select user defined density and heat capacity and add your parameters.

7.1.5 Lumped Battery Settings

1. In the Cell Equilibrium Potential 1 tab, import your OCV-SoC and T-SoC data and fill in the ambient temperature.
2. In the Voltage Losses 1 tab, fill in the ohmic overpotential and activation overpotential values.

7.1.6 Laminar Flow Settings

1. In the Initial Values tab, fill in v_{air} in the expected direction of the airflow.
2. Right-click Laminar Flow and add an Outlet boundary condition, select pressure and fill in 0 static pressure.
3. For the pressure outlet boundary, select the face of the wind tunnel opposite of your desired inlet.
4. Right-click Laminar Flow and add an Inlet boundary condition, select velocity and fill in v_{air} .
5. For the velocity inlet boundaries, select all faces of the wind tunnel (except the cell tabs), that are not pressure outlet.

7.1.7 Heat Transfer in Solids and Fluids Settings

Note that the domain for the active cell material is already created.

1. In the Solid 1 tab, select the current collector and cell tab blocks.
2. In the Fluid 1 tab, select the wind tunnel block.
3. In the Initial Values 1 tab, select and fill in an user defined temperature.
4. Right-click Heat Transfer in Solids and Fluids, add a Dirichlet boundary condition (Temperature) and set the user defined temperature.
5. For this boundary condition, select only the inlet face opposing the outlet (not the side faces).
6. Right-click Heat Transfer in Solids and Fluids, add an Open Boundary condition.
7. For this boundary condition, select only the outlet face.

7.1.8 Mesh

1. Select Mesh 1, choose normal element size.
2. Click Build All.

7.1.9 Study

1. Under Time Dependent Study 1, select Step 1: Time Dependent.
2. Check that all physics and multiphysics modules are active.
3. In the Output Times option, input your desired time range. Use the Range button on the right.
4. In the Study Extensions tab, check Auxiliary Sweep.
5. For Sweep Type, select Specific Combinations.
6. Choose v_{air} from the list under Parameter Name and input your desired range.
7. Click Compute.